

# **IBM University Engagement Project Submission**

*“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”*

**Project Title – [AI Interview Trainer Agent]**

**Domain of Project – [Education / EdTech]**

| Student Name | Recent Passport Photo |                                                                                    | Email ID                | Mobile number [WhatsApp] |
|--------------|-----------------------|------------------------------------------------------------------------------------|-------------------------|--------------------------|
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**Recommended font style and size:**

**Font Style: Arial**

**Font Size (Heading): 28**

**Font Size (Content): 20**

## **Problem Statement -**

**Problem Statement :** AI Interview Trainer Agent

### **The Challenge**

Interview preparation is often generic and repetitive. Candidates may practice questions that are not aligned with their target role, experience level, or technical domain. They also lack consistent, structured feedback on the quality of their answers.

### **The Objective**

Build an AI-powered Interview Trainer Agent that:

- Generates role- and profile-aware interview questions.
- Uses RAG to ground questions and feedback in relevant interview knowledge.
- Evaluates candidate answers for correctness, completeness, clarity, and technical depth.
- Provides actionable feedback, strengths, weaknesses, and improvement suggestions.
- Maintains an interview session and produces a final performance report.

## **Proposed Solution -**

### **Proposed Solution:** AI-Powered Interview Trainer

The proposed solution is an AI-powered Interview Trainer Agent built with FastAPI, IBM watsonx.ai / IBM Granite, RAG, and ChromaDB.

At its core, a Retrieval-Augmented Generation pipeline retrieves relevant interview knowledge and supplies it as context to the Granite model. This helps the system generate more relevant questions and grounded feedback.

Core capabilities:

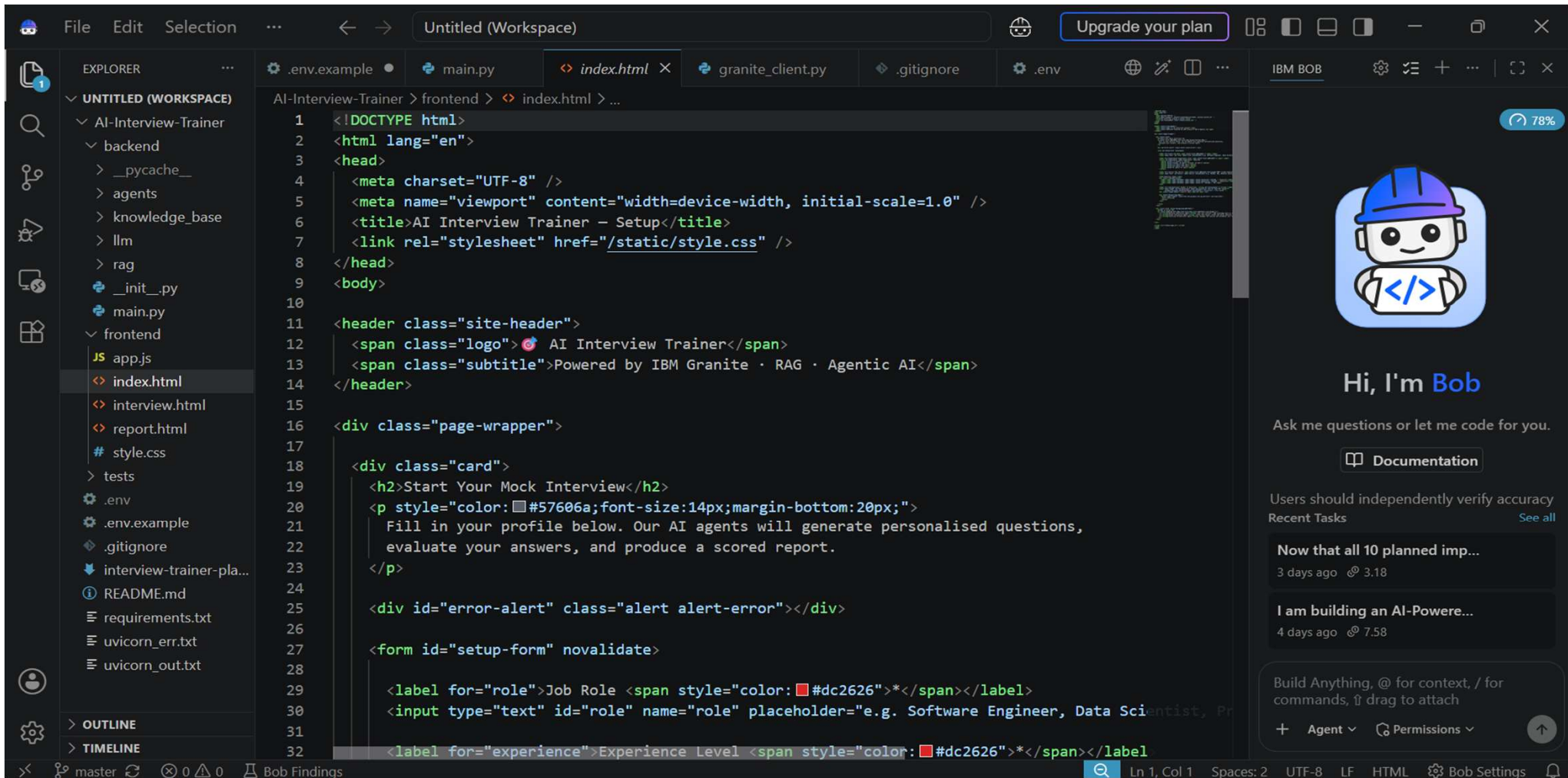
- Candidate profile and target-role aware interview sessions.
- Orchestrated question generation and answer evaluation.
- RAG retrieval from an interview knowledge base.
- IBM Granite for question generation, evaluation, and feedback.
- Session tracking with current-question and final-report APIs.

Overall, the solution acts as a structured mock interviewer rather than a simple chatbot.

## Technology Used

- 1) Python + FastAPI – Backend API and interview-session management.
- 2) IBM Granite – Large language model used for interview question generation, answer evaluation, and feedback.
- 3) IBM watsonx.ai – Cloud platform used to access and run the Granite model.
- 4) RAG (Retrieval-Augmented Generation) – Retrieves relevant interview knowledge before generation.
- 5) ChromaDB – Vector database used to store and retrieve embedded knowledge chunks.
- 6) Sentence Transformers – all-MiniLM-L6-v2 embeddings for semantic retrieval.
- 7) HTML / CSS / JavaScript – Frontend interface for the interview experience.
- 8) Git + GitHub – Version control and public project repository.

## Required Files created using IBM bob -



The screenshot displays a VS Code workspace for an AI Interview Trainer project. The Explorer sidebar on the left shows the project structure, including a 'frontend' directory with files like 'app.js', 'index.html', 'interview.html', 'report.html', and 'style.css'. The Code Editor in the center shows the 'index.html' file, which contains HTML code for a page titled 'AI Interview Trainer - Setup'. The code includes a header with a logo and subtitle, and a main content area with a card for starting a mock interview. The IBM Bob sidebar on the right features a chat interface with a character named Bob, a 'Documentation' link, and a list of recent tasks.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>AI Interview Trainer - Setup</title>
7   <link rel="stylesheet" href="/static/style.css" />
8 </head>
9 <body>
10
11 <header class="site-header">
12   <span class="logo"> AI Interview Trainer</span>
13   <span class="subtitle">Powered by IBM Granite · RAG · Agentic AI</span>
14 </header>
15
16 <div class="page-wrapper">
17
18   <div class="card">
19     <h2>Start Your Mock Interview</h2>
20     <p style="color: #57606a;font-size:14px;margin-bottom:20px;">
21       Fill in your profile below. Our AI agents will generate personalised questions,
22       evaluate your answers, and produce a scored report.
23     </p>
24
25     <div id="error-alert" class="alert alert-error"></div>
26
27     <form id="setup-form" novalidate>
28
29       <label for="role">Job Role <span style="color: #dc2626">*</span></label>
30       <input type="text" id="role" name="role" placeholder="e.g. Software Engineer, Data Scientist, Pr
31
32       <label for="experience">Experience Level <span style="color: #dc2626">*</span></label>
```

# IBM Bob Screenshots:

File Edit Selection ...

← →

Untitled (Workspace)

Upgrade your plan

EXPLORER

▼ UNTIT...  
▼ AI-Interview-Trai...  
  > backend  
  > frontend  
  > tests  
  .env  
  .env.example M  
  .gitignore  
  interview-trainer-pla...  
  README.md  
  requirements.txt  
  uvicorn\_err.txt  
  uvicorn\_out.txt

OUTLINE  
TIMELINE



Show All Commands

Ctrl + Shift + P

Toggle Terminal

Ctrl + `

Go to File

Ctrl + P

IBM BOB



Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

Users should independently verify accuracy

Recent Tasks

Now that all 10 planned imp...  
3 days ago 3.18

I am building an AI-Powere...  
4 days ago 7.58

Build Anything, @ for context, / for commands, ⌘ drag to attach

Agent Permissions

master\*

0 0

Bob Findings

Bob Settings



Ask me questions or let me code for you.

Recent Tasks [See all](#)

I am building an AI-Powere...  
4 days ago 7.58

+ Agent ▾ Permissions ▾

AI-Interview-Trainer > backend > llm >  granite\_client.py

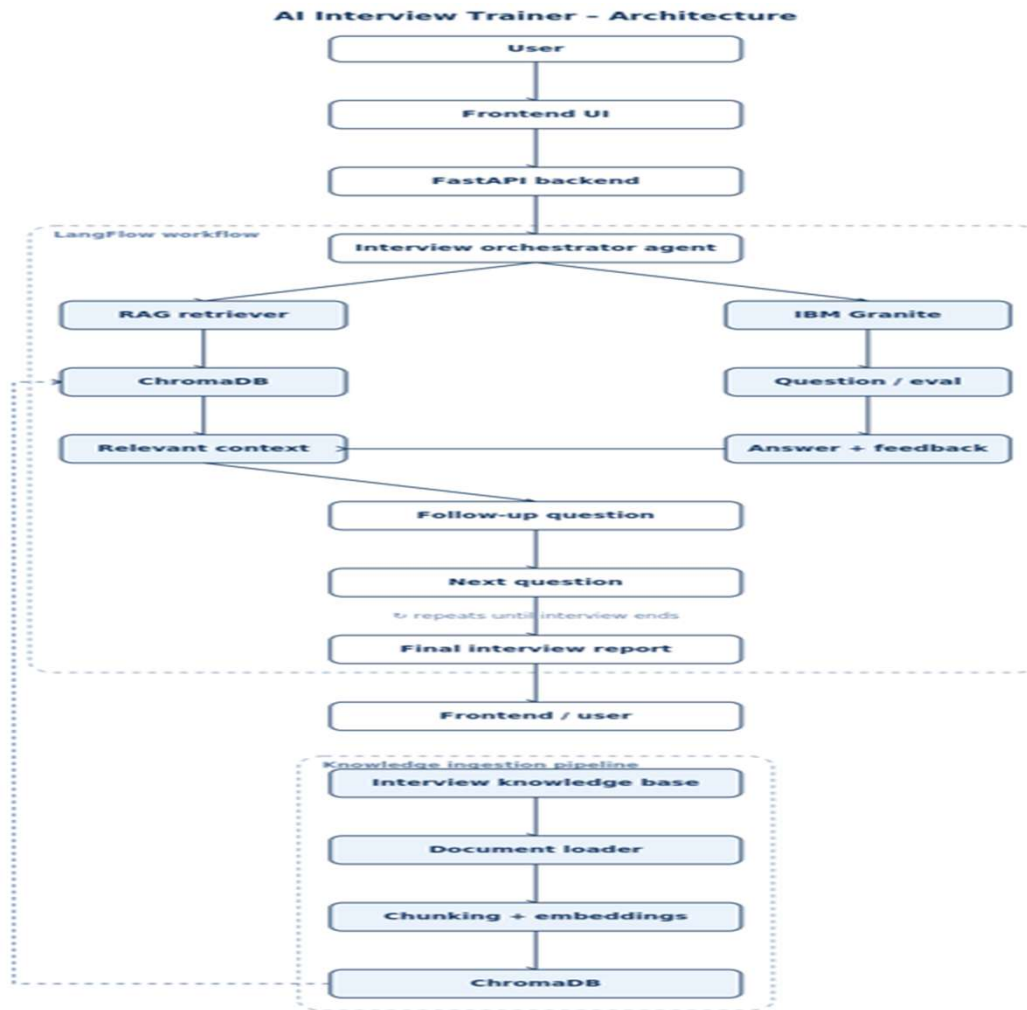
```

10 import os
11 import requests
12 from dotenv import load_dotenv
13
14 load_dotenv()
15
16 # -----
17 # Configuration (read once at import time)
18 # -----
19 _API_KEY = os.getenv("WATSONX_API_KEY", "")
20 _PROJECT_ID = os.getenv("WATSONX_PROJECT_ID", "")
21 _BASE_URL = os.getenv("WATSONX_URL", "https://us-south.ml.cloud.ibm.com")
22
23 _MODEL_ID = "ibm/granite-4-h-small"
24 _IAM_TOKEN_URL = "https://iam.cloud.ibm.com/identity/token"
25 _GENERATE_URL = f"{_BASE_URL}/ml/v1/text/generation?version=2023-05-29"
26
27 # -----
28 # IAM token helper (fetched fresh each call - simple and reliable)
29 # -----
30
31 def _get_iam_token() -> str:
32     """Exchange the API key for a short-lived IAM Bearer token."""
33     if not _API_KEY:
34         raise ValueError(
35             "WATSONX_API_KEY is not set. "
36             "Copy .env.example to .env and fill in your credentials."
37         )
38     resp = requests.post(
39         _IAM_TOKEN_URL,
40         headers={"Content-Type": "application/x-www-form-urlencoded"},
41         data={
42             "grant_type": "apikey",
43             "apikey": _API_KEY,
44             "password": "ibmcloudapikey"
45         }
46     )
47     if resp.status_code != 200:
48         raise ValueError(f"Failed to get IAM token: {resp.status_code}")
49     return resp.json().get("token")

```

 ANUJ KUMAR (13 hours ago)
  Ln 23, Col 35 Spaces: 4 UTF-8 LF Python
  Bob Settings
 

## Architecture blueprint for IBM bob:



# Role of Agentic AI in the solution

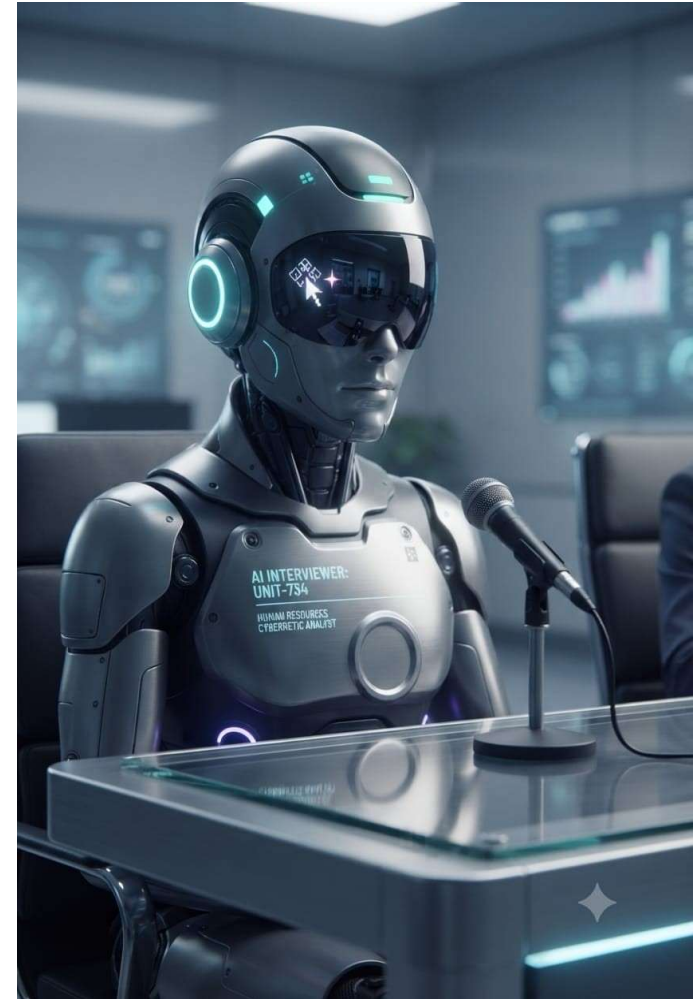
## Role of Agentic AI in AI Interview Trainer

Agentic AI enables the Interview Trainer to act as an intelligent, goal-driven system rather than a simple chatbot. Using IBM watsonx.ai and IBM Granite, the system manages the complete interview process.

The Orchestrator Agent generates relevant interview questions, evaluates candidate answers, provides feedback, and generates follow-up questions based on the candidate's responses.

With RAG and ChromaDB, the agent retrieves relevant interview knowledge and uses it as context while generating questions and evaluating answers.

The agent maintains the interview context and flow, dynamically adapting questions according to the candidate's performance. Overall, Agentic AI makes the system interactive, adaptive, and personalized, providing a realistic mock interview experience.



# Prompt Output Screenshot

AI Interview Trainer — Setup

Ask Gemini

127.0.0.1:8000

AI Interview Trainer

Powered by IBM Granite · RAG · Agentic AI

### Start Your Mock Interview

Fill in your profile below. Our AI agents will generate personalised questions, evaluate your answers, and produce a scored report.

Job Role \*

e.g. Software Engineer, Data Scientist, Product Manager

Experience Level \*

— Select —

Key Skills (comma-separated)

e.g. Python, SQL, Machine Learning, React

Question Types

☒ Technical

☒ Behavioral

☒ HR

Number of Questions: 7

3

10

Start Interview

# Project Output Screenshot

The screenshot shows a web browser window with the title 'AI Interview Trainer — Setup'. The address bar displays '127.0.0.1:8000'. The page features a setup form for an AI interview trainer. The form includes a dropdown menu for selection, a text input for 'Key Skills' with the example 'e.g. Python, SQL, Machine Learning, React', and three checked checkboxes for 'Question Types': 'Technical', 'Behavioral', and 'HR'. A slider for 'Number of Questions' is set to 7, with a range from 3 to 10. A blue 'Start Interview' button is at the bottom of the form. Below the form, a section titled 'How it works' lists three steps: 1. Question Generation Agent, 2. Answer Evaluation Agent, and 3. Interview Orchestrator Agent.

AI Interview Trainer — Setup

Ask Gemini

127.0.0.1:8000

— Select —

Key Skills (comma-separated)

e.g. Python, SQL, Machine Learning, React

Question Types

☒ Technical ☒ Behavioral ☒ HR

Number of Questions: 7

3 10

Start Interview

How it works

1. **Question Generation Agent** uses your profile + RAG knowledge base to create personalised questions via IBM Granite.
2. **Answer Evaluation Agent** scores each answer (1–10) and gives specific improvement suggestions.
3. **Interview Orchestrator Agent** manages the session and assembles your final scored report.

# Project Output Screenshot

AI Interview Trainer — Setup

Ask Gemini

127.0.0.1:8000

### Start Your Mock Interview

Fill in your profile below. Our AI agents will generate personalised questions, evaluate your answers, and produce a scored report.

**Job Role \***

Software Engineer

**Experience Level \***

Fresher (0 years)

**Key Skills** (comma-separated)

C++, react js, node js

**Question Types**

☒ Technical ☒ Behavioral ☒ HR

**Number of Questions: 7**

3 10

**Start Interview**

# Project Output Screenshot

The screenshot displays a web browser window with the address bar showing `127.0.0.1:8000/interview`. The browser's tab is labeled "AI Interview Trainer — Interview". The page header includes the "AI Interview Trainer" logo on the left and the user profile "Software Engineer · Fresher (0 years)" on the right. The main content area features a progress indicator at the top showing "Question 1 of 3" and "0%". Below this, the question is categorized as "Technical" and "Medium" difficulty, labeled "Q1". The question text is: "Write a function in C++ that takes a vector of integers and returns the second largest element without using sorting." Underneath the question is a section titled "Your Answer" containing a large text input field with the placeholder text "Type your answer here...". At the bottom of the answer section is a blue button labeled "Submit Answer".

# Project Output Screenshot

AI Interview Trainer — Interview

Ask Gemini

127.0.0.1:8000/interview

```
// (handles cases where all elements in the vector are equal)
if (secondLargest == INT_MIN) {
    throw std::runtime_error("No unique second largest element exists.");
}

return secondLargest;
}
```

Next Question →

8 / 10

**Excellent!**  
AI Evaluation by IBM Granite

✓ STRENGTHS

- Correct implementation without sorting
- Handles edge cases properly

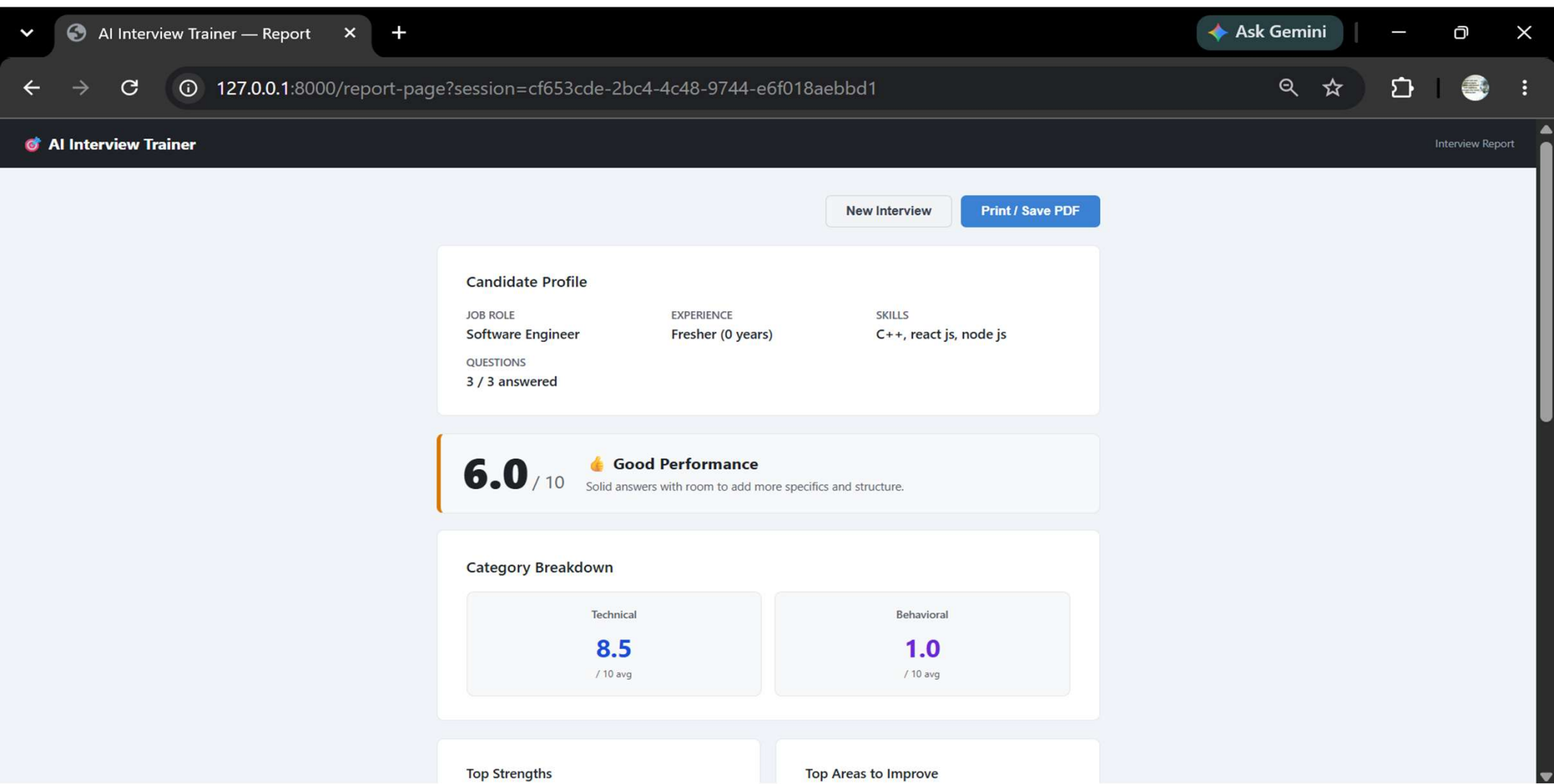
⚡ KEY IMPROVEMENT

- Add comments explaining the logic more clearly

💡 WHAT A STRONG ANSWER INCLUDES

- Explain time complexity as  $O(n)$
- Discuss space complexity as  $O(1)$
- Provide test cases demonstrating correctness

# Project Output Screenshot



# Project Output Screenshot

AI Interview Trainer — Report

Ask Gemini

127.0.0.1:8000/report-page?session=cf653cde-2bc4-4c48-9744-e6f018aebbd1

Category Breakdown

Technical

8.5

/ 10 avg

Behavioral

1.0

/ 10 avg

Top Strengths

- Correct implementation without sorting
- Handles edge cases properly
- Provides complete code implementation

Top Areas to Improve

- Add comments explaining the logic more clearly
- Explain scalability considerations like rooms/channels and horizontal scaling
- Provide an actual response that addresses the question.

Question-by-Question Breakdown

| # | Question                                                                                                                                              | Type   | Score | Key Feedback                                                                                                                                  |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <div>Technical</div> <div>Write a function in C++ that takes a vector of integers and returns the second largest element without using sorting.</div> | Medium | 8/10  | <div>✓ Correct implementation without sorting</div> <div>✗ Add comments explaining the logic more clearly</div>                               |
| 2 | <div>Technical</div> <div>Describe how you would implement real-time updates in a web application using Node.js and web sockets.</div>                | Hard   | 9/10  | <div>✓ Provides complete code implementation</div> <div>✗ Explain scalability considerations like rooms/channels and horizontal scaling</div> |
| 3 | <div>Behavioral</div> <div>Tell me about a challenging project from your studies where you had to learn something new. What was</div>                 | Medium | 1/10  | <div>✓ Answer provided</div> <div>✗ Provide an actual response that addresses the question.</div>                                             |

## **Novelty and Uniqueness**

Adaptive Interview Experience – Generates questions based on job role, experience level, and selected skills.

Agentic Interview Flow – Orchestrates question generation, answer evaluation, feedback, and follow-up questions.

RAG-Based Context – Retrieves relevant interview knowledge from the knowledge base using ChromaDB.

Dynamic Follow-Up – Adapts the next question according to the candidate's previous answer.

Multi-Type Assessment – Supports Technical, Behavioral, and HR interview questions.

Personalized Evaluation – Provides candidate-specific feedback and a final scored interview report.

IBM Granite Powered – Uses IBM Granite through watsonx.ai for intelligent question generation and evaluation.

Overall: Makes mock interviews interactive, adaptive, and personalized rather than a fixed question-answer system.

## Git Hub Link

- Please create public github repository (Enable Add README button while creating repository) with format –  
Please test and Paste the working Github URL –**<https://github.com/anujcoder3480/AI-Interview-Trainer>**

**Make sure you have Uploaded following three files into your github repository**

- 1) All Agent relevant files of IBM Cloud , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

## **Future Scope**

### **1. Voice-Based Interviews –**

Enable voice interaction so candidates can answer questions naturally through speech.

### **2. Resume-Based Personalization –**

Integrate resume analysis to generate interview questions based on the candidate's projects, skills, and experience.

### **3. Advanced Performance Analytics –**

Provide detailed analysis of technical knowledge, communication, confidence, and answer quality with performance trends.

### **4. Multilingual Interviews –**

Support multiple languages to make the interview trainer accessible to a wider range of candidates.

## Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)



## Certificates Earned:

Add your IBM BOB Certificate

| IBM SkillsBuild                                                                   | Completion Certificate         |
|-----------------------------------------------------------------------------------|--------------------------------|
|  |                                |
| This certificate is presented to                                                  |                                |
| Anuj Kumar                                                                        |                                |
| for the completion of                                                             |                                |
| <b>Lab: Troubleshoot Your Code Using IBM Bob</b>                                  |                                |
| (ALM-COURSE_4071307)                                                              |                                |
| According to the Adobe Learning Manager system of record                          |                                |
| <b>Completion date:</b> 11 Sep 2026 (GMT)                                         | <b>Learning hours:</b> 30 mins |

# Thank You!

Thank you for your time and interest.